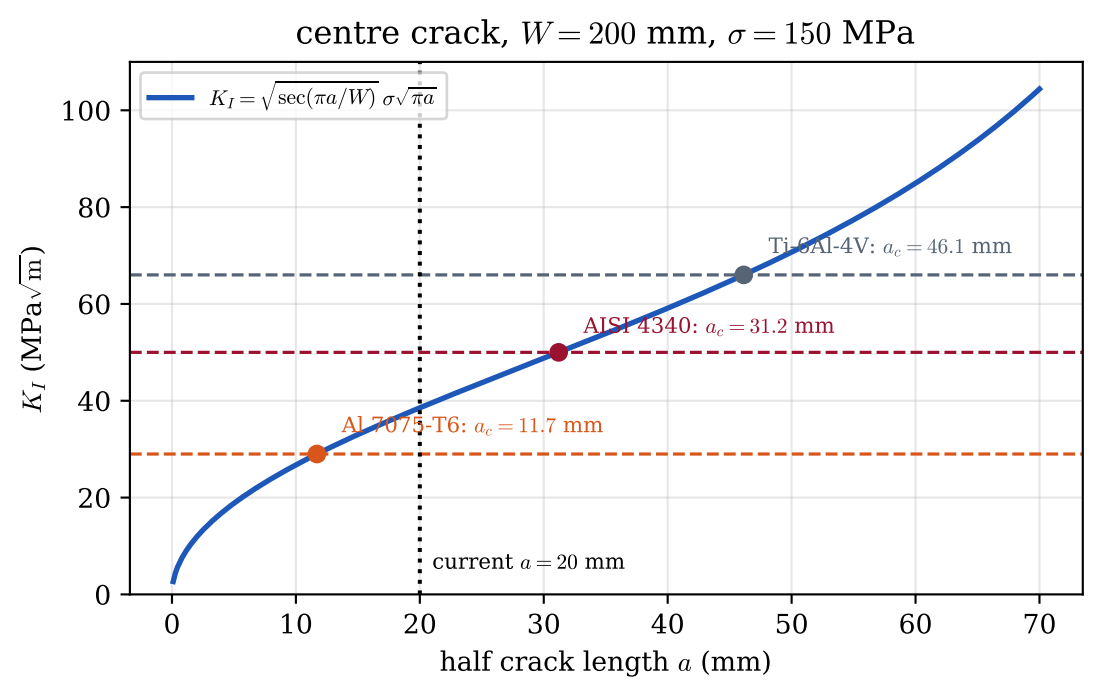
centre crack, $W = 200$ mm, $\sigma = 150$ MPa
$K_I = \sqrt{\sec(\pi a/W)}\,\sigma\sqrt{\pi a}$
Ti-6Al-4V: $a_c = 46.1$ mm
AISI 4340: $a_c = 31.2$ mm
Al 7075-T6: $a_c = 11.7$ mm
current $a = 20$ mm
K_I (MPa$\sqrt{\text{m}}$)
half crack length a (mm)
0
20
40
60
80
100
0
10
20
30
40
50
60
70